

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Hackathon	Assessment:	Assessment Tool 3– Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	YEDURURI ANJANI PRIYA	Reg. No.:	192324176
Section / Batch:	2023/AI&DS	Date:	31.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Problem statement:

Disease Symptom Checker. Predict likely disease category from a patients symptom vector (fever, cough, fatigue, etc.) using kNN with different values of k and distance metrics (Euclidean vs Manhattan).

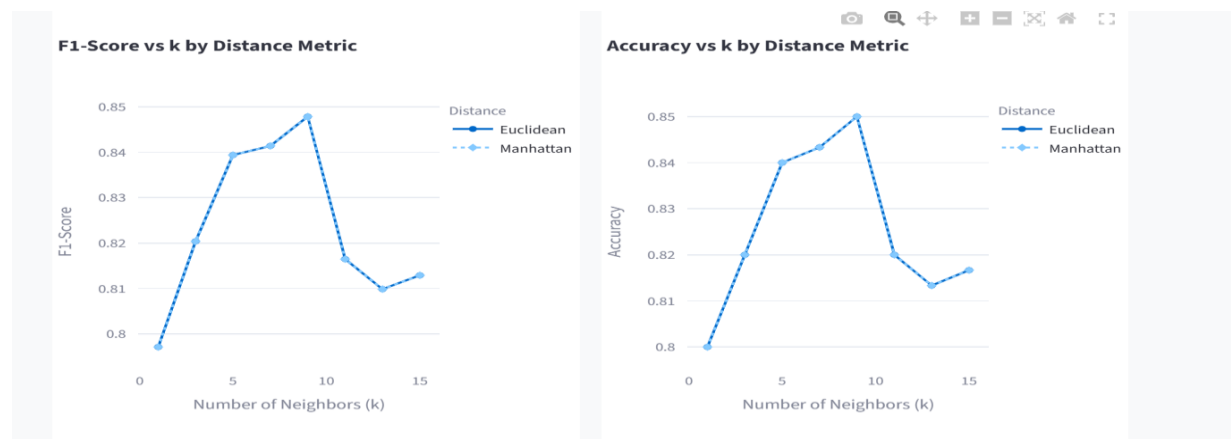
Python code:

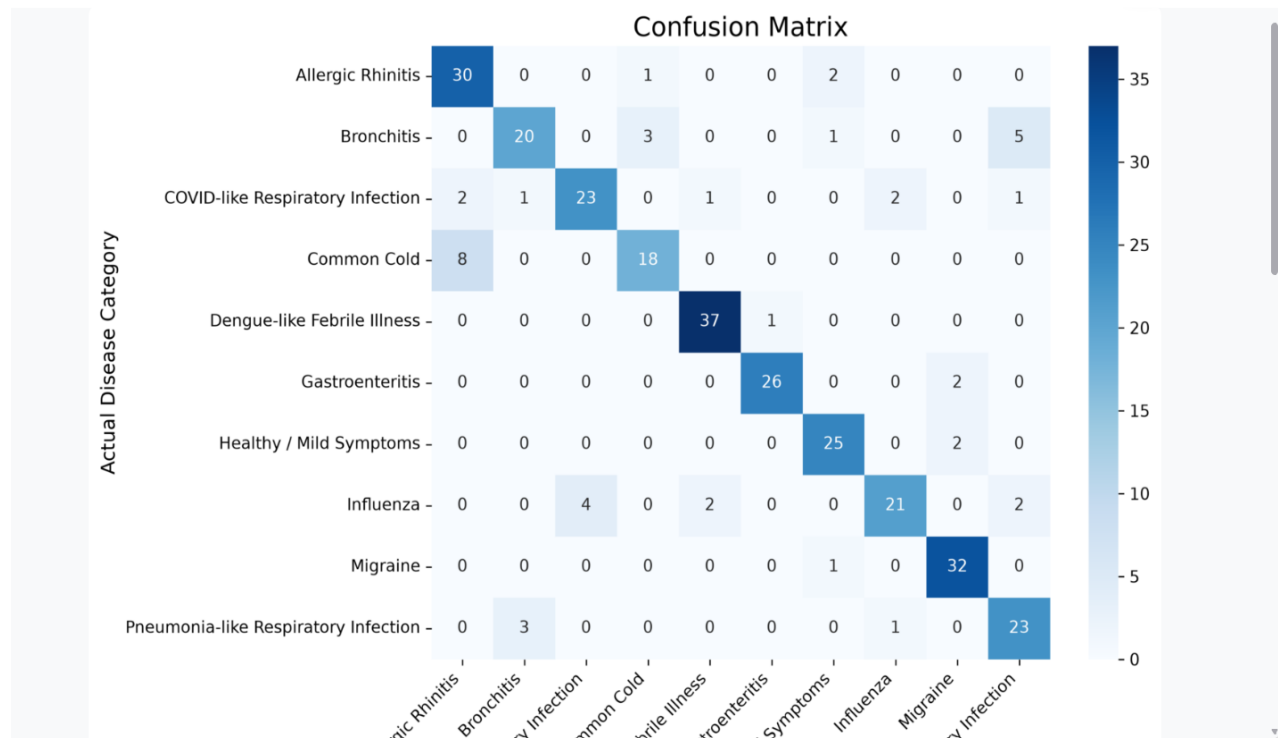
```
> data
> results
> src
◆ .gitignore
◆ app.py
① README.md
≡ requirements.txt

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"""
st.markdown("### How it works")
workflow = go.Figure(data=[go.Sankey(
    node = dict(
        pad = 15,
        thickness = 20,
        line = dict(color = "black", width = 0.5),
        label = ["Patient Symptoms", "Symptom Vector (0",
        color = ["#3B82F6", "#10B981", "#8B5CF6",
    ),
    link = dict(
        source = [0, 1, 2, 3, 4],
        target = [1, 2, 3, 4, 5],
        value = [1, 1, 1, 1, 1]
    )
))]
workflow.update_layout(height=400)
st.plotly_chart(workflow, use_container_width=True)
```

Output





Geotag

